

MATH 340

Quiz 2

February 7, 2025

1. (10 points) Problem 1

- (a) A first order ODE is written as $P(x, y)dx + Q(x, y)dy = 0$. Write down the condition for the ODE to be exact.
- (b) Consider the ODE $(e^x y^2 + \sin y)dx + (2e^x y + x \cos y)dy = 0$. Determine whether it is exact, show your work.
- (c) For the ODE in part (ii), if it is exact, find the general solution.

2. (10 points) Problem 2. Consider the population model $y'(t) = \frac{y}{10} \left(1 - \frac{y}{K}\right)$ with capacity $K > 0$.

- (a) Find the general solution of the ODE.
- (b) Assume the initial population is one-third of the capacity. Find the time at which the population has doubled.
- (c) Sketch the solution in Part (ii) on the ty -plane. Find the limit $\lim_{t \rightarrow \infty} y(t)$.